Taha Koleilat

PhD Candidate
Applied Scientist Intern at Amazon

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My Webpage
Google Scholar
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Education

2024–Present **PhD, Electrical & Computer Engineering**, *Concordia University*, Montreal, Canada.
Research focus: computer vision, vision-language models, multimodal learning, foundation models, and medical image analysis.

2023–2024 **Master of Applied Science, Electrical & Computer Engineering**, *Concordia University*, Montreal, Canada.
CGPA: 4.00/4.30. Coursework includes Medical Image Processing, Machine Learning, Biological Signal Processing, and Deep Learning.

2019–2023 **Bachelor of Engineering, Computer & Communications Engineering**, *American University of Beirut*, Beirut, Lebanon.
CGPA: 4.00/4.00; High Distinction. **Minor:** Economics. Coursework includes Machine Learning, Computer Networks, Security, Software Engineering, and Embedded Systems.

Experience

Applied Scientist Intern, Amazon

Jun 2026 – Present ***Amodal, Real-Time 3D Orientation Estimation in Videos***, *Amazon*, Canada.

- Research and develop occlusion-aware methods for amodal 3D orientation estimation, targeting accurate real-time inference in complex video scenes.
- Design and benchmark computer vision models under partial occlusion, substantial viewpoint changes, dynamic motion, and temporal variation.
- Build a large-scale video dataset and annotation pipeline, including data collection, quality assurance, and coverage of diverse orientations and occlusion patterns.
- Optimize model architectures and inference pipelines to improve accuracy, latency, and computational efficiency.

Graduate Research Assistant, IMPACT and Health-X Labs

Sep 2023 – Present ***Foundation Models and Multimodal Learning for Biomedical Applications***, *Concordia University*, Montreal, Canada.

- Develop biomedical vision-language models for multimodal representation learning, image classification, and text-guided medical image segmentation.
- Design data- and parameter-efficient adaptation methods that generalize across datasets, imaging modalities, clinical domains, and downstream tasks.
- Build reproducible training and evaluation pipelines for large-scale experimentation, benchmarking, and ablation studies.
- Conduct research under the supervision of **Dr. Hassan Rivaz** and **Dr. Yiming Xiao**, resulting in publications at leading computer vision and medical imaging venues.

Artificial Intelligence Engineer, Reality AI Lab

Apr 2024 – Jan 2025 ***LLM-Powered Tools for Education and Career Development***, *Reality AI Lab*.

- Designed and developed AI-powered educational products using large language models from OpenAI and Google Gemini.

- Developed ReX, a personalized AI career coach that provides learners with tailored guidance, mentorship, and support throughout their career journey.
- Expanded Kai, an AI educator that processes user-provided documents to generate summaries, multiple-choice questions, and structured course syllabi.

Research Intern, Trusted Networks Lab

May 2022 – Oct 2022

Cyber-Social Consensus for Supply-Chain Decision-Making, *Trusted Networks Lab*.

- Developed a cyber-social consensus decentralized application using Web3.py, Tkinter, email services, and Solidity smart contracts.
- Integrated email-based stakeholder participation with smart contracts to automate voting and decision-making in supply-chain scenarios.
- Improved the transparency, fairness, and reliability of multi-stakeholder decisions under the supervision of **Dr. Raja Jurdak**.

Publications

- **Taha Koleilat**, Hojat Asgariandehkordi, Omid Nejati Manzari, Berardino Barile, Yiming Xiao, and Hassan Rivaz, "MedCLIPSeg: Probabilistic Vision–Language Adaptation for Data-Efficient and Generalizable Medical Image Segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 [**Paper**][**Page**].
- **Taha Koleilat**, Hojat Asgariandehkordi, Hassan Rivaz, and Yiming Xiao, "BiomedCoOp: Learning to Prompt for Biomedical Vision-Language Models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [**Paper**][**Page**].
- Omid Nejati Manzari, Hojat Asgariandehkordi, **Taha Koleilat**, Yiming Xiao, and Hassan Rivaz, "Sparse Spectral LoRA: Routed Experts for Medical VLMs," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026 [**Paper**][**Page**].
- **Taha Koleilat**, Hassan Rivaz, and Yiming Xiao, "Evi-Steer: Learning to Steer Biomedical Vision-Language Models through Efficient and Generalizable Evidential Tuning," in *29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026, **Early accept (Top 9%)** [**Paper**][**Page**].
- **Taha Koleilat**, Hojat Asgariandehkordi, Hassan Rivaz, and Yiming Xiao,"MedCLIP-SAM: Bridging Text and Image Towards Universal Medical Image Segmentation," in *27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024, (Acceptance rate: $\simeq$ 30%)[**Paper**][**Page**].
- **Taha Koleilat**, Hojat Asgariandehkordi, Hassan Rivaz, and Yiming Xiao,"MedCLIP-SAMv2: Towards Universal Text-Driven Medical Image Segmentation", *Medical Image Analysis*, 2025 [**Paper**][**Page**].
- **Taha Koleilat**, Hassan Rivaz, and Yiming Xiao,"Singular Value Few-shot Adaptation of Vision-Language Models", *TMLR*, 2026 [**Paper**][**Page**].
- **Taha Koleilat***, Pascal Spiegeler*, Arash Harirpoush, Corey S. Miller, Hassan Rivaz, Marta Kersten-Oertel, Yiming Xiao, "TextSAM-EUS: Text Prompt Learning for SAM to Accurately Segment Pancreatic Tumor in Endoscopic Ultrasound", *the Third ICCV Workshop on Computer Vision for Automated Medical Diagnosis (CVAMD)*, 2025 [**Paper**][**Code**].
- Hamza Rasaee, **Taha Koleilat**, Hassan Rivaz,"GroundingDINO-US-SAM: Text-Prompted Multi-Organ Segmentation in Ultrasound with LoRA-Tuned Vision-Language Models", *IEEE TUFFC*, 2025 [**Paper**][**Code**].
- Omid Nejati Manzari, Hojat Asgariandehkordi, **Taha Koleilat**, Yiming Xiao, and Hassan Rivaz,"Medical Image Classification with KAN-Integrated Transformers and Dilated Neighborhood Attention", *Applied Soft Computing*, 2025 [**Paper**][**Code**].

Honors and Awards

2026 Recipient of the best poster award at the 10th GSR conference at Concordia University.

2025 Recipient of the ***FRQNT Doctoral Training Scholarship***, awarded after being ranked 2nd by the reviewing committee. This scholarship supports my research on transforming healthcare through universal, efficient, and scalable Vision-Language Models (VLMs) (Value: $100,000 CAD).

2025 Recipient of the ***Concordia University Doctoral Graduate Fellowship***, awarded to exceptional doctoral students for outstanding research and academic excellence (Value: 28,000$ CAD).

2024 – 2025 Received two ***Gold Level IEEE TMI Distinguished Reviewer Certificate*** *as an acknowledgment of my significant contribution to the journal.*

2024 Recipient of the ***International Tuition Award of Excellence*** *which reduces the tuition to the Quebec rate* (Value: 47,463$ CAD)

2023 Recipient of the ***Concordia Merit Scholarship*** *entrance award for Master's students* (Value: 5,000$ CAD)

2023 Graduated from the American University of Beirut with ***High Distinction***

2023 Received the ***Dean's Award for Creative Achievement*** *for our project titled "COVID-19 Indoor Access Rules Verification using ML" which was placed 1st among 30 other groups in Computer Engineering.*

2021 Recipient of ***MEPI-TLS Scholarship*** awarded by AUB Tomorrow's Leaders Gender Scholars Program (Value: 13,110$ USD)

2020 – 2023 Recipient of ***A.M. Rabbat Endowed Scholarship*** (Value: 5,244$ USD), ***Dr. Saad Hamdi AlZaim and Family Scholarship*** (Value: 9,911$ USD), and ***Nabil Zuhair Haddad Scholarship*** (Value: 24,822$ USD) to provide additional financial support to qualified students who excel academically pursuing degrees in Engineering

2019 – 2023 Placed on the ***Dean's Honor list*** for all semesters attended at the American University of Beirut.

Computer skills

Languages Python, Java, C/C++, SQL, JavaScript, Solidity, R, LaTeX

Frameworks PyTorch, TensorFlow, Django, Flask

Tools Git, Docker, Google Cloud Platform, Overleaf, Visual Studio, PyCharm, IntelliJ, Eclipse, Excel

Libraries OpenCV, Scikit-learn, pandas, NumPy, Matplotlib

Position of Responsibility

2024-Present **Reviewer for IEEE TMI and TPAMI, MICCAI, MedIA, Nature Medicine**.

2024-present **Student member of IEEE**.

2023-2026 **Student member of Quebec Bio-imaging Network (QBIN)**.

Teaching

COEN366: Communication Networks & Protocols, Concordia University.

COMP233: Probability & Statistics, Concordia University.

COMP248: Object Oriented Programming I, Concordia University.

COMP478/6771: Image Processing, Concordia University.

ELEC366: Telecommunication Networks, Concordia University.

EECE311: Electronic Circuits, American University of Beirut.